# Mohanad HUSSEIN



+49 15510 880734 m.hussein@stud.uni-goettingen.de Göttingen, Germany
Mohanad Hussein github.com/mohanadhussein

## Summary

Bioinformatics Master's student with strong command of Python, R, Matlab & Bash. Pursuing a career in the intersection of OMICS integration in diagnostic applications and cloud infrastructure utilization. Experienced in employing scientific software & HPC resources for both research and clinical purposes, in addition to wet-lab methods for molecular genetics and genomic database curation. Mainly interested in developing tools for impactful Human Genomics research.

## Education

**MSc Computational Biology & Bioinformatics** University of Göttingen (GA) **Göttingen, Germany** *2024-2026*

Core Courses: Bioinformatics Algorithms, Statistics in R, Deep Learning (CV), ML in R, Python & Data Science.

**BSc Molecular Biology & Genetics** Istanbul Technical University (İTÜ) **Istanbul, Türkiye** *2020-2024*

Core Courses: Human & molecular Genetics, Probability & Stats, Biochemistry, R programming, MATLAB & Python.

## Areas of Expertise

Workflow Development & Management - Image Processing - Machine Learning - Containerization - Version Control - High Performance Computing - NGS Data Analysis - Genetic Linkage Analysis - DNA Isolation & Purification

## Experience

**Project Intern** Max Planck Institute for Multidisciplinary Sciences **Göttingen, Germany** *04/2026 - present*

- Developing a phylogenetic tree-reconstruction benchmark workflow in Snakemake based on the OMA database (in XML, JSON, nwk) developed by the Swiss Institute of Bioinformatics, to test Soeding Lab in-house tool prior to publishing.

**Working Student** GWDG - AG Computing **Göttingen, Germany** *10/2025 - present*

- Assisting in testing & monitoring Bioinformatics tools and datasets on the GWDG HPC cluster.
- Developing tutorials & materials for using Bioinformatics tools and datasets as part of the GWDG HPC Life Science services.

**Project Intern** Auditory Neuroscience Institute (@DPZ) **Göttingen, Germany** *08/2025 - 12/2025*

- Gene mapping in Middle Eastern and European Families for Genetic Hearing Loss by Linkage & NGS Analysis.
- Established a pipeline to streamline Analysis of sample data.

**Project Intern** UniFr/Swiss Institute of Bioinformatics **Fribourg, Switzerland** *07/2024 - 09/2024*

- Performed primary-to-tertiary analysis of low-depth NGS data using ATLAS tool developed in Snakemake by Wegmann's group.
- Established a pipeline in R to test & optimize in-lab tool "BeXY" that identifies sex-linked scaffolds through Bayesian Inference.

**Lab Assistant** MOBGAM Research Center **Istanbul, Türkey** *03/2023 - 06/2024*

- Coordinated and secured TÜBİTAK 2209-A funding (Turkish Government) to sustain and finalize a pandemic-impacted research project on *VWA1* in Prof. Aslıhan Tolun's Human Genetics Lab.
- Handled isolation and preparation of 200 DNA samples followed by pre-sequencing analysis & quality control.
- Conducted Whole Exome Sequencing (WES) analysis and Fine Mapping through Linkage Analysis.
- Reviewed Sanger, WGS & WES results and human genetic variation databases (ClinVar, dbSNP, GenomeAD) prior to publishing.
- Established the pipeline for Variant Calling and R-based Linkage analysis in the lab.
- Co-authored a paper & presented a scientific poster of the project at the ESHG 2026 (Gothenberg, Sweden).

**Project Intern** EPFL - Brain Mind Institute **Lausanne, Switzerland** *07/2023 - 09/2023*

- Investigated the impact of genetic variation on brain structure & morphology in *Dmel-DGRP* lines using GWAS and GSEA.
- Established Docker containers-based pipeline to perform Gene Set Enrichment Analysis (GSEA) and presented results in R shiny.
- Prepared Micro-computed Tomography-scanned brains followed by image analysis and segmentation using DragonFly software.
- Presented a poster at the Swiss Drosophila Meeting 2023 in Fribourg.

**Research Intern** Lund University **Lund, Sweden** *07/2023 - 08/2022*

- Designed and performed an evolutionary experiment for reproductive success between Balancer X-chromosome & WT *Dmel* lines.
- Completed a Population Genetics summer course at the GENECO research school.
- Assisted in maintaining *D. Melanogaster* and *M. Lignona* cultures.

## External Projects & Contributions

**ENVIRONMENTAL NICHE MODELLING IN R** Eurasia Institute of Earth Science *10/2023 - 01/2024*

Performed Environmental Niche Modelling (ENM) analysis for the invasive Raccoon Dog species (suspected of Covid-19 spread) in Turkey to predict it's future distribution given different Shared Socioeconomic Pathway (SSP) using 'BioMod' package in R, concluding that 5 main climatic variables indicate it's limited expansion in Northeastern Turkey.

**NEUROSCIENCE SUMMER PROGRAM** IYNA *06/2022 - 08/2022*

International Youth Neuroscience Association (IYNA) neuroscience program final project of scientific paper review and poster presentation on "Altered Gene Expression in Neurons during Programmed Cell Death".

**VizJobs** GitLab Repository *04/2025 - 06/2025*

Led the development of a python-based command line tool for listing job vacancies in Germany, utilizing API of the official government platform for employment, Linkedin and similar platforms to estimate vacancies' salaries using supervised machine learning trained on proprietary database accessed through an industrial collaborator.

## Awards & Scholarships

- Swiss European Mobility Fellowship (July 2024 - Sep. 2024)
- TÜBITAK 2209-A annual project fund (Sep. 2023 - Sep. 2024)
- EPFL funded summer internship (July 2023 - Sep. 2023)
- Erasmus+ summer Fellowship (July 2022 - Aug. 2022)
- IsD Bank's 5 yrs international scholarship program upon academic performance (Sep. 2019 - Sep 2024)

## Publications & Posters

Önel, E., **Hussein, M.**, Nalbant, G., Sönmez, E., Mutlu, P., Aksan-Kurnaz, I., Yıldız-Bölükbaşı, E., Malik, S., Tolun, A. (2026) Heterozygous variants of *VWA1* isoform 2 in Zygodactyly 1. **Journal of Human Genetics**. (Research Article Under Review) & **European Society of Human Genetics 2026**

**Hussein, M.**, Katrine K., Abbott, J. (2023) Assessing Sperm Competition Between *Drosophila Melanogaster* lines. **Swiss Drosophila Meeting 2023**

## Certifications

- GWDG: Shiny, Jupyter, Parallel Programming in MPI
- Uni-Göttingen: NGS Analysis of RNAseq & ChIPseq data
- EPFL: Genomics & Bioinformatics (Python & R)
- WELLCOME: Bioinformatics for Biologists
- INRIA: Machine learning in Python with scikit-learn
- DataCamp: Version Control with Git/Github

## Languages

- **English** C1
- **Turkish** C1
- **Arabic** Native
- **German** A2

## References

- **Prof. Dr. Aslıhan Tolun**
  tolunasli@itu.edu.tr
- **Dr. Ana Marija Jaksic**
  ana.jaksic@epfl.ch
- **Prof. Dr. Vona Barbara**
  barbara.vona@med.uni-goettingen.de
- **Prof. Dr. Daniel Wegmann**
  daniel.wegmann@unifr.ch
- **Prof. Dr. Jessica Abbott**
  jessica.abbott@biol.lu.se